$$\angle PSR + \angle PQR = 180^\circ \quad \text{(state reason_____)}$$
$$70^\circ + \angle PQR = 180^\circ$$
$$\angle PQR = 180^\circ - 70^\circ$$
$$\angle PQR = 110^\circ$$

In $\Delta PQR$ we have,

$$\angle PQR + \angle PRQ + \angle QPR = 180^\circ \quad \text{(state reason_____)}$$
$$110^\circ + \angle PRQ + 40^\circ = 180^\circ$$
$$\angle PRQ = 180^\circ - 150^\circ$$
$$\angle PRQ = 30^\circ$$

## Exterior Angle of a Cyclic Quadrilateral

An exterior angle of a quadrilateral is an angle in its exterior formed by one of its sides and the extension of an adjacent side.

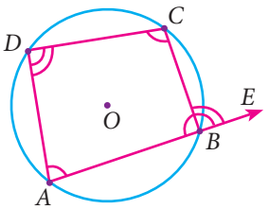


Fig. 4.82

Let the side $AB$ of the cyclic quadrilateral $ABCD$ be extended to $E$. Here $\angle ABC$ and $\angle CBE$ are linear pair, their sum is $180^\circ$ and the angles $\angle ABC$ and $\angle ADC$ are the opposite angles of a cyclic quadrilateral, and their sum is also $180^\circ$. From this, $\angle ABC + \angle CBE = \angle ABC + \angle ADC$ and finally we get $\angle CBE = \angle ADC$. Similarly it can be proved for other angles.

**Theorem 13** If one side of a cyclic quadrilateral is produced then the exterior angle is equal to the interior opposite angle.

### Progress Check

1. If a pair of opposite angles of a quadrilateral is supplementary, then the quadrilateral is _____.
2. As the length of the chord decreases, the distance from the centre _____.
3. If one side of a cyclic quadrilateral is produced then the exterior angle is _____ to the interior opposite angle.
4. Opposite angles of a cyclic quadrilateral are _____.

### Example 4.11

In the figure given, find the value of $x^\circ$ and $y^\circ$.

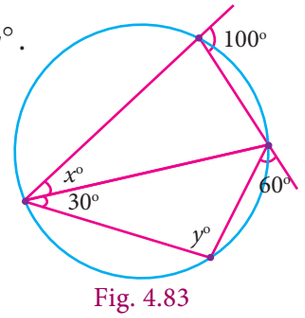


Fig. 4.83

*Solution*

By the exterior angle property of a cyclic quadrilateral,

we get, $y^\circ = 100^\circ$ and

$x^\circ + 30^\circ = 60^\circ$ and so $x^\circ = 30^\circ$



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